CS/SE 4AL3 Homework 2: Classification with SVMs

Fall 2026
Due: October 20th
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1 Overview

This assignment will get you started working with support vector machine classifiers and evaluation for classification problems. You will build a model to predict solar flare events. You will experiment with real solar activity data through a combination of three feature sets and evaluate with cross validation and confusion matrices.

2 Instructions

This assignment has one part and will be graded out of 25 points: 16 points for implementation, 6 points for answers to the questions, and 3 points for code readability and organization. You must work on this assignment individually.

You may use ONLY the following libraries in your code. You will receive a zero if any other libraries are used besides the ones listed below. Note that scikit learn is available for this assignment. You may add import statements within these packages for your convenience, e.g. `from sklearn.PACKAGE import CLASS`.

Listing 1: Available Packages

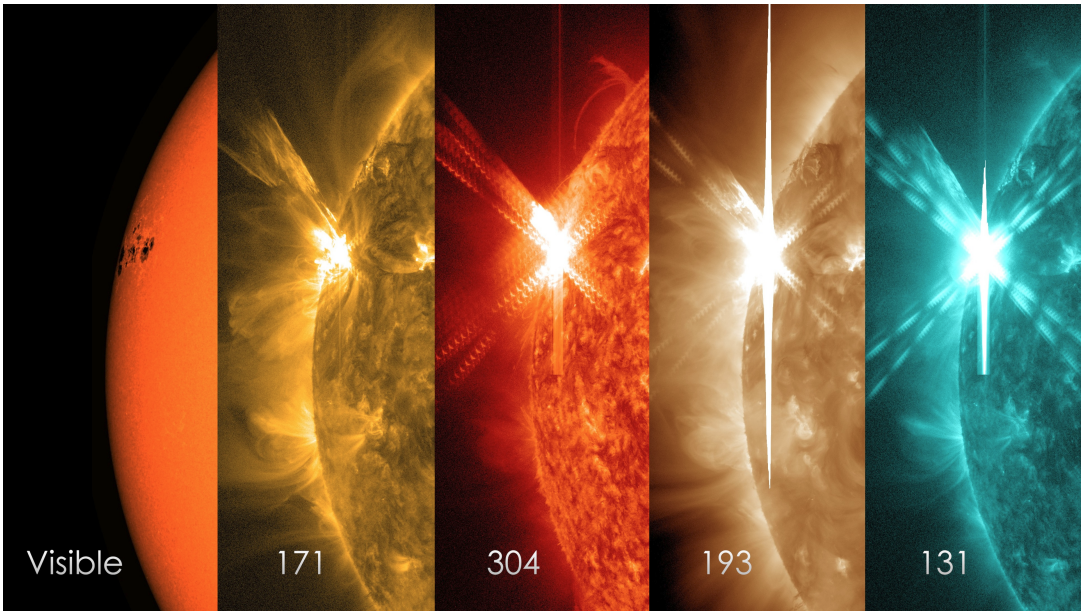
```
import os, sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn
import sklearn
import random
```

Your code should be written in Python and should include enough documentation/instructions for someone else to be able to run. You must submit your work as a Jupyter notebook (.ipynb). Make sure that your file read/write operations use UTF-8 (if relevant). Your code should run with Python version 3.12. You should fill in the sections of the code that say "#CODE HERE".

You are welcome to use code snippets from examples in class, things you find online, or from AI code generation tools. Just make sure to give proper attribution to code you did not write. Follow the syllabus instructions for how to report the use of AI tools. However, you may not copy code that does the entire assignment (e.g. someone who did this assignment in a previous semester).

3 Problem

Context: The sun's solar flares are large bursts of electromagnetic energy that can significantly impact life on Earth. Powerful solar flares can knock out power grids, satellite systems, and all communication systems on the planet. To monitor solar activity and study such events, the United States National Oceanic and Atmospheric Administration (NOAA) operates the Geostationary Operational Environmental Satellite System; a series of geosynchronous satellites with specialized measurement instruments onboard. The GOES system provides us with solar imagery, magnetometer data, solar X-ray data, and data on high-energy solar protons that hit Earth. This data



is sent to Earth and regularly updated on the GOES server which is open for public use.

Unlike Earth, the sun's regions are not divided by countries, states, or cities. Instead, various patches on the sun are numbered by NOAA for scientific investigation purposes, and the most active patches are frequently monitored for high-intensity bursts of electromagnetic radiation. Each patch is called a HARP (HMI Active Region Patch); an enduring, coherent magnetic structure that produces an electromagnetic field. The regions provide measurable features that characterize that patch. There are two classes of solar flare events of particular interest to scientists: the M-class and X-class. Solar flares occur in HARP regions and the level of the energy burst is measured.

Monitoring these patches is helpful, but it is more useful to predict when a powerful burst will occur. Predicting an upcoming solar flare 24 hours in advance can give us time to mitigate damage. This is very challenging because solar flares are rare events. Most importantly, we do not know which features directly indicate an upcoming solar flare.

Challenge: In this assignment, you will build a classfier for binary classification using data from the Helioseismic and Magnetic Imager Instrument on NASA's Solar Dynamics Observatory (SDO) satellite that captures various solar events. It is the first instrument that continuously maps the vector magnetic field of the sun. The magnetic activity recorded using these instruments will serve as the feature set for the model you build.

A major solar event is defined as a burst of GOES X-ray flux of peak magnitude above the M1.0 level. A *positive solar event* is defined as an active HARP region that flares with a peak magnitude above the M1.0 level, as reported in the GOES database. A *negative solar event* is an active region that does not have such an event (no flare above M1.0 level) within 24 hours. The goal of the classification model should be to train on the given data and predict whether a major solar event will occur in the next 24 hours. This means that the classifier must predict whether a given solar event is positive (indicating a flaring active region) or negative (indicating a non-flaring region).

For this assignment, you will implement a Support Vector Machine (SVM) classifier that can distinguish between a positive and a negative solar flar event. Scientists M.G. Bobra and S. Couvidat used the SVM to study solar flares and published their findings in 2015. You may be interested to read the paper, however, it is optional and all information you need for the assignment has already been provided. You will evaluate your classifier using 5-fold cross-validation on the provided dataset.

Feature Name	Description	Column Number and Filename
HARPNUM NOAA_ARS	HARP Number of the sun patch Corresponding NOAA assignments	Column 1-2: all_harps_with_noaa_ars.txt
USFLUX MEANGAM MEANGBT MEANGBZ MEANGBH MEANJZD TOTUSJZ MEANALP MEANJZH TOTUSJH ABSNJZH SAVNCPP MEANPOT TOTPOT MEANSHR SHRGT45 R_VALUE AREA_ACR	Total unsigned flux Mean angle of field from radial Mean gradient of total field Mean gradient of vertical field Mean gradient of the horizontal field Mean vertical current density Total unsigned vertical current Mean characteristic twist parameter Mean current helicity Total unsigned vertical current Absolute value of the net current helicity Sum of modulus of the net current per polarity Mean photospheric magnetic free energy Total photospheric magnetic free energy density Mean shear angle Fraction of area with shear >45 degrees Sum of flux near polarity inversion line Area of strong field pixels in the active region	Column 1-18: Main feature set (FS -I) pos_features_main_timechange.npy neg_features_main_timechange.npy
Time-Change Features	Changes in the values of the above 18 properties every 6 hours for 24 hours prior to the time we make a prediction (4 times)	Column 19-90: feature set (FS -II) pos_features_main_timechange.npy neg_features_main_timechange.npy
MaxMin Features	The difference between maximum and minimum values in the 24 hours prior to prediction time for each feature	Column 1-18: Max Min Feature (FS -IV) pos_features_maxmin.npy neg_features_maxmin.npy

Dataset: The dataset is gathered from the GOES data server using SunPy, an open-source software for solar physics. The data is directly accessed from the server, providing a neat interface for data structures and methods to query. The provided dataset in `data-2010-15` records solar activity from May 1, 2010 until May 1, 2015. Use the three feature sets FS-I, FS-II, and FS-IV listed below. The original feature-set names are kept to match the data files. The following table shows the features of the dataset and the files in which they are saved.

The data folder contains only the four feature files used in this assignment:

- `pos_features_main_timechange.npy`
- `neg_features_main_timechange.npy`
- `pos_features_maxmin.npy`
- `neg_features_maxmin.npy`

The notebook provides code to load these files. The main time-change files contain FS-I and FS-II, and the max-min files contain FS-IV. Positive observations receive label 1, and negative observations receive label 0. No additional data files are needed.

Goals (16 points): Use the provided dataset and feature files. Your goal is to create an input data array, provide appropriate labels, implement the SVM model, and evaluate it. Please implement functions that:

1. **Preprocess the data (3 points)** to prepare it for the model by:

(a) **Normalizing the features (1 point).** Use `StandardScaler` to standardize each feature. Fit the scaler on the training rows only within each cross-validation fold, then use the same scaler to transform the test rows. Do not fit the scaler on the full dataset before cross-validation.

(b) **Removing missing values, if any (1 point).** Check all three included feature sets together. If an observation has a missing or infinite value, remove that observation from every feature set and its label. Use the same retained observations for every feature combination. Do not remove valid zero values.

(c) **Assigning appropriate labels (1 point).** Combine the positive and negative observations from the provided dataset, using label 1 for positive observations and 0 for negative observations. Keep the labels aligned with the feature rows.

2. **Create the input data array (2 points)** by concatenating selected features in a single 2D array. Write this function such that you can input any combination of the three feature sets and create the input data. There are **7 combinations**: (FS-I), (FS-II), (FS-IV), (FS-I, FS-II), (FS-I, FS-IV), (FS-II, FS-IV), and (FS-I, FS-II, FS-IV). Award 1 point for selecting the correct features and 1 point for combining them correctly without changing the observation order.

3. **Perform classification using a Support Vector Machine (2 points).** First evaluate all 7 feature combinations using the same baseline settings: `C=1.0`, `gamma='scale'`, `kernel='rbf'`, and `class_weight=None`. Award 1 point for correctly training the SVM and 1 point for evaluating all 7 combinations at these settings and identifying the best combination by mean TSS. If combinations are tied, report the ties and choose the one with fewer columns; if still tied, use the order listed above.

4. **Perform 5-fold cross-validation (4 points).** Use `StratifiedKFold` with `n_splits=5`, `shuffle=True`, and `random_state=42`. Reuse the same folds for all feature combinations. Award 1 point for correct fold construction, 1 point for training on the training rows and predicting only the test rows, 1 point for reporting all five TSS scores and their mean and standard deviation (`ddof=0`), and 1 point for storing one test-fold prediction per observation and constructing the combined confusion matrix. Report the results separately for each feature combination.

5. **Calculate the true skill score (TSS) (2 points)** using the following equation: $TSS = \frac{TP}{TP+FN} - \frac{FP}{FP+TN}$. TSS is a good measure to predict rare events when we have a large class imbalance in the input data. Use this as the performance metric for all models. Award 1 point for the correct confusion-matrix counts and 1 point for the correct TSS calculation. Use the class order $[0, 1]$ so the matrix is arranged as $[[TN, FP], [FN, TP]]$.

6. **Tune the hyperparameters (3 points).** Use the best feature combination from the baseline experiment. Use `GridSearchCV` with TSS as the scoring metric and the same five folds. Choose the values of `C`, `gamma`, and the kernels you want to test, and briefly explain your choices. Gamma does not affect a linear kernel, so avoid repeating identical linear models for different gamma values. Put `StandardScaler` and `SVC` in a `Pipeline` so scaling is fitted within each training fold during the search. Keep `class_weight=None`. Award 1 point for a suitable search and a clear explanation of the chosen settings, 1 point for using TSS and training-only scaling with the same folds, and 1 point for reporting the best settings and their mean and standard deviation of CV TSS alongside the baseline result for the same feature combination. Keep the baseline results unchanged.

4 Deliverables

Submit one completed **Jupyter notebook (.ipynb)** and notebook PDF on Avenue. It must contain your code, outputs, plots, tables, answers to the questions, setup/run instructions, and AI disclosure. Do not submit a Python file, a separate PDF report, or any datasets. Restart the kernel and run all cells before submitting.

Your notebook must include:

1. Bar charts showing the TSS of each baseline feature combination across all five folds. Label the axes and feature combinations clearly. A multi-panel figure is acceptable.

2. A confusion matrix for each of the 7 baseline feature combinations. Use the combined test-fold predictions, not training predictions. Label the actual and predicted classes clearly.

3. A table with all five fold scores, mean TSS, and standard deviation for each baseline feature combination. Add a separate baseline-versus-tuned comparison for the selected combination, including the settings, mean TSS, and standard deviation. These tables and plots support your answers; they do not receive separate marks.

4. **Answers to the following questions (6 points):**

   (a) **(2 points)** Which feature combination worked best and which was worst in the baseline experiment? Report their mean TSS and standard deviation, including any ties.

   (b) **(2 points)** Does adding additional feature sets improve the TSS score? Support your answer with at least two comparisons from your baseline results.

   (c) **(2 points)** Which hyperparameter settings gave the highest mean CV TSS for the selected feature combination? Compare the mean TSS and standard deviation before and after tuning. How did tuning impact your score? How do you expect it to perform on new data? Explain your answer.

5. **AI disclosure:** Complete the AI disclosure section in the notebook, including a statement if no AI tools were used. Follow the syllabus requirements for what to report.

Code readability and organization (3 points):

- **1 point:** Use clear variable and function names, consistent indentation, and readable formatting.

- **1 point:** Add short, useful comments or docstrings explaining the main steps and function inputs and outputs.

- **1 point:** Organize the code in a logical order and provide clear setup and run instructions in the notebook. The notebook should run from start to finish with the supplied data.

Total: 16 points for implementation + 6 points for answers to the questions + 3 points for readability and organization = 25 points.

5 Help

Please use the Teams channel and tutorials to ask questions about the assignment when you need guidance or pointers on this homework. You are free to discuss your approach and ideas with classmates, but should not share code or reuse data. You may use generative AI tools if you find them helpful, but you must clearly document how they were used and follow the guidelines in the syllabus. If you use generative AI and do not report it, you may receive a 0 for the assignment. You take full responsibility for the deliverables you submit.